MSG No. 131.435 $P4_2/mmc$ [Type I, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `mmm`.

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,4,5,6,9,12,13,14]`
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[2,3,7,8,10,11,15,16]`

Table 2: Wyckoff site: `2b`, site symmetry: `mmm`.

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[1,4,5,6,9,12,13,14]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[2,3,7,8,10,11,15,16]`

Table 3: Wyckoff site: `2c`, site symmetry: `mmm`.

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,4,5,6,9,12,13,14]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[2,3,7,8,10,11,15,16]`

Table 4: Wyckoff site: `2d`, site symmetry: `mmm`.

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[1,4,5,6,9,12,13,14]`
2	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[2,3,7,8,10,11,15,16]`

Table 5: Wyckoff site: `2e`, site symmetry: `-4m2`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,6,7,8,10,11,12,13]`
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[2,3,4,5,9,14,15,16]`

Table 6: Wyckoff site: `2f`, site symmetry: `-4m2`

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,6,7,8,10,11,12,13]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[2,3,4,5,9,14,15,16]`

Table 7: Wyckoff site: `4g`, site symmetry: `2mm`.

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,6,12,13]`
2	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[2,3,15,16]`

continued ...

Table 7

No.	position	mapping
3	$[0,\ 0,\ -z]$	`[4,5,9,14]`
4	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[7,8,10,11]`

Table 8: Wyckoff site: `4h`, site symmetry: `2mm`.

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[1,6,12,13]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[2,3,15,16]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[4,5,9,14]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[7,8,10,11]`

Table 9: Wyckoff site: `4i`, site symmetry: `2mm`.

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,6,12,13]`
2	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[2,3,15,16]`
3	$\left[0,\ \frac{1}{2},\ -z\right]$	`[4,5,9,14]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	`[7,8,10,11]`

Table 10: Wyckoff site: `4j`, site symmetry: `m2m`.

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,4,13,14]`
2	$\left[0,\ x,\ \frac{1}{2}\right]$	`[2,7,11,16]`
3	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[3,8,10,15]`
4	$[-x,\ 0,\ 0]$	`[5,6,9,12]`

Table 11: Wyckoff site: `4k`, site symmetry: `m2m`.

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[1,4,13,14]`
2	$\left[\frac{1}{2},\ x,\ 0\right]$	`[2,7,11,16]`
3	$\left[\frac{1}{2},\ -x,\ 0\right]$	`[3,8,10,15]`
4	$\left[-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[5,6,9,12]`

Table 12: Wyckoff site: `4l`, site symmetry: `m2m`.

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[1,4,13,14]`
2	$[0,\ x,\ 0]$	`[2,7,11,16]`
3	$[0,\ -x,\ 0]$	`[3,8,10,15]`

continued ...

Table 12

No.	position	mapping
4	$\left[-x,\, 0,\, \frac{1}{2}\right]$	[5,6,9,12]

Table 13: Wyckoff site: 4m, site symmetry: m2m.

No.	position	mapping
1	$\left[x,\, \frac{1}{2},\, 0\right]$	[1,4,13,14]
2	$\left[\frac{1}{2},\, x,\, \frac{1}{2}\right]$	[2,7,11,16]
3	$\left[\frac{1}{2},\, -x,\, \frac{1}{2}\right]$	[3,8,10,15]
4	$\left[-x,\, \frac{1}{2},\, 0\right]$	[5,6,9,12]

Table 14: Wyckoff site: 8n, site symmetry: ..2

No.	position	mapping
1	$\left[x,\, x,\, \frac{1}{4}\right]$	[1,7]
2	$\left[-x,\, x,\, \frac{3}{4}\right]$	[2,5]
3	$\left[x,\, -x,\, \frac{3}{4}\right]$	[3,4]
4	$\left[-x,\, -x,\, \frac{1}{4}\right]$	[6,8]
5	$\left[-x,\, -x,\, \frac{3}{4}\right]$	[9,15]
6	$\left[x,\, -x,\, \frac{1}{4}\right]$	[10,13]
7	$\left[-x,\, x,\, \frac{1}{4}\right]$	[11,12]
8	$\left[x,\, x,\, \frac{3}{4}\right]$	[14,16]

Table 15: Wyckoff site: 8o, site symmetry: .m.

No.	position	mapping
1	$[0,\, y,\, z]$	[1,12]
2	$\left[-y,\, 0,\, z+\frac{1}{2}\right]$	[2,15]
3	$\left[y,\, 0,\, z+\frac{1}{2}\right]$	[3,16]
4	$[0,\, -y,\, -z]$	[4,9]
5	$[0,\, y,\, -z]$	[5,14]
6	$[0,\, -y,\, z]$	[6,13]
7	$\left[y,\, 0,\, \frac{1}{2}-z\right]$	[7,10]
8	$\left[-y,\, 0,\, \frac{1}{2}-z\right]$	[8,11]

Table 16: Wyckoff site: 8p, site symmetry: .m.

No.	position	mapping
1	$\left[\frac{1}{2},\, y,\, z\right]$	[1,12]
2	$\left[-y,\, \frac{1}{2},\, z+\frac{1}{2}\right]$	[2,15]
3	$\left[y,\, \frac{1}{2},\, z+\frac{1}{2}\right]$	[3,16]
4	$\left[\frac{1}{2},\, -y,\, -z\right]$	[4,9]
5	$\left[\frac{1}{2},\, y,\, -z\right]$	[5,14]

continued ...

Table 16

No.	position	mapping
6	$\left[\frac{1}{2}, -y, z\right]$	[6,13]
7	$\left[y, \frac{1}{2}, \frac{1}{2} - z\right]$	[7,10]
8	$\left[-y, \frac{1}{2}, \frac{1}{2} - z\right]$	[8,11]

Table 17: Wyckoff site: 8q, site symmetry: m..

No.	position	mapping
1	$[x, y, 0]$	[1,14]
2	$\left[-y, x, \frac{1}{2}\right]$	[2,11]
3	$\left[y, -x, \frac{1}{2}\right]$	[3,10]
4	$[x, -y, 0]$	[4,13]
5	$[-x, y, 0]$	[5,12]
6	$[-x, -y, 0]$	[6,9]
7	$\left[y, x, \frac{1}{2}\right]$	[7,16]
8	$\left[-y, -x, \frac{1}{2}\right]$	[8,15]

Table 18: Wyckoff site: 16r, site symmetry: 1

No.	position	mapping
1	$[x, y, z]$	[1]
2	$\left[-y, x, z + \frac{1}{2}\right]$	[2]
3	$\left[y, -x, z + \frac{1}{2}\right]$	[3]
4	$[x, -y, -z]$	[4]
5	$[-x, y, -z]$	[5]
6	$[-x, -y, z]$	[6]
7	$\left[y, x, \frac{1}{2} - z\right]$	[7]
8	$\left[-y, -x, \frac{1}{2} - z\right]$	[8]
9	$[-x, -y, -z]$	[9]
10	$\left[y, -x, \frac{1}{2} - z\right]$	[10]
11	$\left[-y, x, \frac{1}{2} - z\right]$	[11]
12	$[-x, y, z]$	[12]
13	$[x, -y, z]$	[13]
14	$[x, y, -z]$	[14]
15	$\left[-y, -x, z + \frac{1}{2}\right]$	[15]
16	$\left[y, x, z + \frac{1}{2}\right]$	[16]